1. (30 points) Answer the following questions briefly.
   (a) Increasing the supply voltage of a CMOS circuit can boost the operation speed of the CMOS circuit. However, the designer typically doesn't use this approach. Why? (4 points)
   (b) List the process for transferring the mask pattern to the surface of a silicon region. (3 points)
   (c) Explain how to tune the size of FinFET transistors. (2 points)
   (d) List two features of static CMOS circuits. (2 points)
   (e) A CMOS circuit consumes static power caused by two static currents. List the two static currents. (2 points)
   (f) List the steps of chip-on-wafer assembly process for integrating a die on an interposer. (5 points)
   (g) Explain the back side power supply technology and list its two advantages. (4 points)
   (h) In a CMOS gate, the transistors connected to the gate output suffer from the severest body effect. Why? (2 points)
   (i) Switching speed of a MOS transistor depends on multiple parameters. List the parameters. (3 points)
   (j) The output capacitance of a CMOS logic gate in a CMOS circuit is comprised of three components. List the three components. (3 points)

2. (15 points) Fig. 1 shows a multiplexer-based circuit. (a) Complete the circuit such that it realizes the boolean functions $f(x,y,z) = xy + yz$ and $f(x,y,z) = x + y + z$ when $s = 1$ and $s = 0$, respectively. (10 points) (b) If the 2-to-1 multiplexer is realized by transmission gates without buffer insertion, express the delay of the circuit in terms of the number of multiplexer stages. Explain your answer(points)

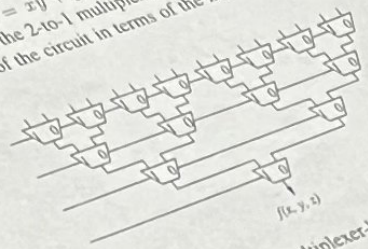


Figure 1: A multiplexer-based circuit

3. (10 points) Fig. 2 shows a CMOS circuit driving a load capacitance $C_L = C$. The inverter A is a minimum-size inverter (the size of NMOS and PMOS is W/L) and its input capacitance is C_g. Assumptions: 1) the drain and wire parasitic capacitance can be neglected; 2) $e = 2.7$, $e^2 = 7.2$, $e^3 = 20$, and $e^4 = 54$; 3) $\mu_n = \mu_p$; 4) the turn-on resistance of a PMOS transistor with W/L is R. If the simple RC model is used to estimate the delay and the output rise time t_r of the NAND gate D is $0.05RC$, design the buffer shown in dash line such that the delay T is minimized.

4. (10 points) Fig. 3(a) shows a CMOS logic gate. Assume that the threshold voltage of the NMOS transistor is 0.4V; the turn on resistance of N1, N2, and N3 is R, 2R, and R, respectively. If the initial voltage of the output Vout is 1.2V and an input stimulus (0→1.2V) as shown in Fig. 3(b) is applied, how much voltage of the input stimulus (Vin) at z is? (4 points) Identify the operation region of the transistors N1, N2, and N3 from t1 to t2. (6 points)

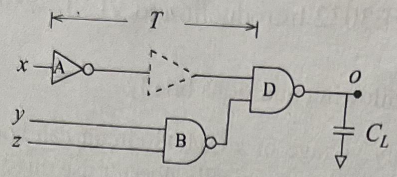


Figure 2: A simple CMOS circuit.

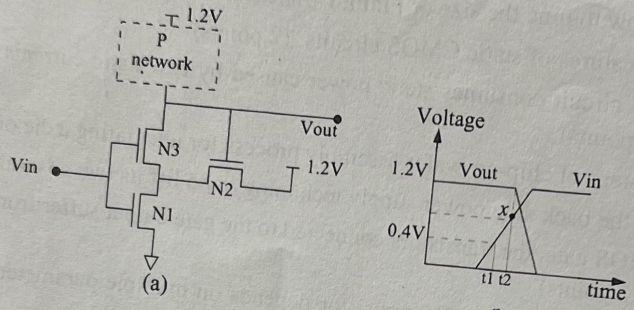


Figure 3: (a) A CMOS logic gate (b) waveforms of Vin and Vout.

5. (10 points) Fig. 4(a) shows a CMOS circuit. Assume that the sizes of NMOS and PMOS transistors in the logic gates are W/L and 2W/L, respectively. Also, $\mu_n = 2\mu_p$. If the functional inputs of the circuits are (ABCD)={0011, 0101, 1101, 1001}, explain the relationship of noise margins NM_L and NM_H (i.e., $NM_L > NM_H$, $NM_L < NM_H$, or $NM_L = NM_H$) of the gates G2 and G3.

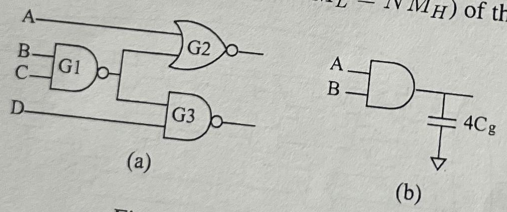


Figure 4: CMOS circuits

6. (10 points) For a given technology, the turn-on resistance of a W/L NMOS is R and $\mu_n = 2\mu_p$. Also, the gate capacitance of a W/L NMOS or PMOS transistor is C_g. Fig. 4(b) shows a two-input AND gate drives a 4 C_g capacitive load. Design the AND gate such that the worst-case delay of the AND gate is smaller than $8RC_g$, where the delay is estimated by using simple RC model and the parasitic capacitance can be ignored. Draw the transistor-level schematic diagram and label the size of each transistor in terms of W/L.

7. (15 points) Realize the following Boolean functions using CMOS logic design style. (a) $F = \overline{X(Y+Z)}$ (5 points) (b) $F = \overline{XY+Z}$ (5 points) (c) $F = \overline{X+Y(W+Z)}$ (5 points)